

ARKA PATRA

CYBER SECURITY STUDENT — B.SC, GNIT • Security-Focused Developer • Python • Web Security

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PROFESSIONAL PROFILE: I build practical security-focused applications, web tools, and developer projects, exploring real cybersecurity concepts through working code. My work focuses on web security, phishing detection, password security, security analysis, privacy-oriented tools, and practical software development.

CAREER OBJECTIVE: Motivated Cyber Security student seeking an opportunity to work with an organization that offers challenges, responsibilities, and decision-making opportunities to build knowledge, technical experience, and contribute to organizational success.

TECHNICAL SKILLS

- **Programming:** Python, JavaScript, HTML, CSS, SQL
- **Development:** Flask, SQLite, Frontend Development, Responsive UI, REST / API concepts
- **Tools:** VS Code, Git, GitHub, Microsoft Office
- **Cybersecurity:** Web Security, Phishing Awareness, Password Security, Security Analysis, Cybersecurity Fundamentals
- **Operating Systems:** Windows 11, Linux
- **Soft Skills:** Problem Solving, Communication, Teamwork, Logical Thinking, Continuous Learning

KEY TECHNICAL PROJECTS

Website Security Copilot | Cybersecurity • Browser Extension

AI-powered assistant that scans, explains, and helps fix website security issues — plus an Always-On browser guard.

- Designed as a browser add-on; live demo shows the interface, while full protection starts after extension install.
- One-click website check examines safety settings, site lock, cookies, open doors, and email safety.
- Always-On Guard watches every page opened and marks it Safe, Doubtful, or Blocked automatically.
- Displays seriousness, supporting proof, simple explanations, ready fixes, and an A–F grade for detected issues.
- Provides a simple 0–100 safety score with clear actionable fixes for non-technical users.
- Manual scan engine checks security headers (CSP, HSTS, X-Frame-Options), SSL expiry, cookie flags, and phishing signals.

Technical Stack / Details: Next.js, Tailwind CSS, FastAPI, Python, dnspython, SQLite, Docker, Manifest V3 Extension.

Altron Password Inspector | Cybersecurity • Password Security

A simple tool that creates strong passwords and tells users how safe they are.

- Generates truly random strong passwords using browser-side cryptographic randomness (`crypto.getRandomValues()`).
- Allows users to choose password length, letters, numbers, and symbols with show, copy, and clear controls.
- Evaluates and displays live password strength as weak, medium, or strong based on 15+ char threshold and character classes.
- Works fully client-side in the browser ensuring credentials are never transmitted externally.

Technical Stack / Details: JavaScript, HTML, CSS, Web Crypto API.

URL Threat Scanner | Web Security • Phishing Detection

Paste a link and receive a 0–100 danger score with a clear safe-or-not answer.

- Detects suspicious URL patterns including raw IP addresses in place of domains and short hidden links.
- Identifies copy-cat brand names and common fraud / phishing words.
- Displays LOW RISK, MEDIUM RISK, or HIGH RISK results with a visual risk indicator.
- Uses client-side pattern analysis without requiring a server for core URL inspection.

Technical Stack / Details: JavaScript, HTML, CSS; checks raw IP hosts, uncommon TLDs, lookalike brand spelling.

NexVault (Working) | Cybersecurity • Data Protection • Group Project

Personal secure locker web application designed around authentication and private data access.

- Users create an account and log in before entering their private locker page.
- Stores private notes and data behind authentication and access control following 'your data, your vault, your control'.
- Currently marked In Progress and identified on portfolio as a collaborative group project.

Technical Stack / Details: Web App, Authentication, Data Security, Vault Dashboard; encrypted storage concept.

TECHNICAL PROJECTS (CONTINUED)

Humanize AI (Working) | Web Development • AI Text Tool

Turns robot-sounding AI writing into more natural human writing.

- Provides three transformation choices: Light, Medium, and Strong with word count change tracking.
- Includes an offline-capable browser fallback so the tool functions without internet connectivity.
- Uses a React frontend with Express backend while offering a zero-build client-side fallback for static hosting.

Technical Stack / Details: React, Node.js, Express, JavaScript; offline fallback and real-time transformation.

Hidden-Prompt Scanner for Documents | Cybersecurity • Prompt Injection Detection

Finds hidden messages in documents that may attempt to manipulate or fool AI systems.

- Scans PDF, DOCX, TXT, MD, and HTML content for hidden prompt-injection indicators and zero-width characters.
- Detects white-on-white text, extremely small text, transparent text, off-page content, and file metadata notes.
- Matches suspicious phrases (e.g., 'ignore previous instructions') and calculates overall risk (LOW, MEDIUM, HIGH).

Technical Stack / Details: Python, Flask, PyPDF2, python-docx, BeautifulSoup; deployed on Render.

Prototype Safety Check | Cybersecurity • Prototype Safety

Passive health and security check for a deployed website link, returning Safe, Risky, or Danger.

- Performs six gentle passive safety checks covering database errors, bad files, fake pages, and security locks.
- Checks exposed parameters, missing headers, suspicious scripts/iframes, HTTPS signals, rate-limit, and HSTS.
- Provides a clean web UI, CLI scanner, and GET API endpoint storing previous checks for review.

Technical Stack / Details: Python, Flask, HTML, JavaScript, REST API, Render.

Laptop Antivirus | Cybersecurity • Real-Time Protection

Always-on desktop protection concept that monitors apps, sites, images, videos, and files.

- Supports automatic startup protection with manual ON/OFF toggle and automatic re-enable behavior.
- Monitors apps, websites, USB drives, and media in real-time with threat severity alerts and Quarantine/Delete actions.
- Includes ransomware, USB, phishing, and gaming-mode shields with an encrypted quarantine vault using SHA-256 signatures.

Technical Stack / Details: Python standard library, Tkinter, file watcher, SHA-256 signatures, URL shield, quarantine vault.

INTERNSHIP EXPERIENCE

Thiranex — Skill Development & Future Tech | Cyber Security Intern | AUG 2026 — SEPT 2026

Certificate ID: THX-AUG0426-563

- Completed a Cyber Security internship with Thiranex (MSME-registered), involving hands-on training and applied project work under industry mentorship.
- Participated in practical cybersecurity learning and applied project activities as part of the internship experience.
- Worked under industry mentorship to strengthen practical cybersecurity knowledge and understand core security concepts through applied work.

EDUCATION

B.Sc. in Cyber Security

2024 – Present

Guru Nanak Institute of Technology (GNIT), MAKAUT, Kolkata, India • Sem 1: Pending | Sem 2: 8.09 | Sem 3: 7.29 | Sem 4: 7.35

Higher Secondary (Class XII)

2024

Madhyamgram APC Vidyayatan, WBBSE • Aggregate Score: 63.2%

Secondary (Class X)

2022

Madhyamgram APC Vidyayatan, WBBSE • Aggregate Score: 60.71%

CERTIFICATIONS & JOB SIMULATIONS

• Mastering Technical Interviews (Forge (JUL 2026))

Mock technical interviews — problem-solving, algorithm & data-structure

• Cyber Job Simulation (Deloitte · Forge (JUL 2026))

Real-world cyber tasks in a simulated Deloitte engagement — security

challenges.

- **Introduction to Cybersecurity** (Cisco Networking Academy (JUL 2025))

Core security concepts — threats, vulnerabilities, and defense frameworks.

- **Intro to Cybersecurity Simulation** (Commonwealth Bank · Forage (JUL 2026))

Bank-grade security drills covering incident response, penetration testing & analysis.

analysis & response.

- **Computer Hardware Basics** (Cisco Networking Academy (JUN 2025))

PC architecture, hardware maintenance, device security, and safety standards.

- **Partnering with AI in Workplace** (Datacom · Forage (JUL 2026))

AI-assisted research, prompt engineering, automated workflows, and debugging.

PORTFOLIO & PROFILES

• **Portfolio:** <https://portfolio-ten-gold-16.vercel.app/> • **GitHub:** <https://github.com/arkap1502> • **LinkedIn:** <https://www.linkedin.com/in/arka-patra-579110422/>